SIMATS ENGINEERING
CO2_AT1_Scenario-Based Assessment

NAME: C Prasanna **COURSE NAME: Software Testing**

REG NO:192371035 **COURSE CODE: CSA3**

COURSE OUTCOME COVERED

CO2: Design effective test cases using black-box, white-box testing techniques and to control methods flow.
Assessment Tool Weightage: 50%

Sl. No	Questions	Bloom's Level
1	An online banking system is experiencing frequent failures during fund transfers, especially at boundary values (e.g., minimum and maximum transfer limits). As a test engineer, critically analyze the scenario and evaluate the current test case design strategy. Apply appropriate black box testing techniques such as equivalence partitioning and boundary value analysis to redesign the test cases. Assess the effectiveness of random testing used by the team and justify whether it should be retained or replaced. Propose a structured and optimized testing approach ensuring maximum defect detection. Present your solution with clear reasoning and justification.	5
2	A healthcare application module has undergone testing, but critical defects were found post-deployment due to insufficient coverage. Evaluate the testing approach used and analyze gaps in test adequacy criteria. Apply white box testing techniques by constructing a conceptual control flow graph and identifying missing paths. Critically assess the coverage achieved (statement, branch, or path coverage). Design an improved testing strategy ensuring complete coverage and justify your decisions. Explain how your approach enhances reliability and reduces risk.	5
3	A retail e-commerce system uses multiple input fields (price, quantity, discount), but testers failed to detect errors in specific input combinations. Analyze the scenario and evaluate the limitations of the current black box testing approach. Apply appropriate techniques such as equivalence class partitioning, boundary value analysis, and other black box methods (e.g., decision table testing). Propose a revised test design strategy covering all possible input combinations. Justify your approach in terms of improved defect detection and test completeness. Provide a structured explanation of your solution.	5
4	A software module has complex nested conditions and loops, but the testing team reports only partial path coverage. Evaluate the current white box testing approach and identify gaps in control flow graph construction and path testing. Analyze how inadequate identification of	5

	independent paths impacts defect detection. Propose a complete path testing strategy ensuring all critical paths are tested. Justify your decisions using coverage criteria and expected outcomes. Present your analysis in a clear and structured manner.	
5	A financial application combines both black box and white box testing, yet defects continue to appear in production. Critically analyze the integration of test case design strategies used in the project. Evaluate whether test adequacy criteria and coverage metrics were properly applied. Propose an improved hybrid testing strategy combining black box techniques (ECP, BVA) and white box methods (path testing, CFG). Justify how your approach ensures better coverage and defect prevention. Conclude with a well-structured and reasoned solution.	5
6	A specialized HR payroll system calculates monthly bonuses based on years of service and performance ratings. Testers found that employees at exactly 5, 10, and 15 years were receiving incorrect payouts. Critically analyze the failure and evaluate the current test design. Apply Boundary Value Analysis (BVA) and Equivalence Class Partitioning (ECP) to redesign the test suite. Justify why these techniques are superior to exploratory testing for this specific financial logic.	5
7	An automated temperature control system for a laboratory uses complex if-else logic to manage heaters and coolers. Post-release, the system failed to trigger the alarm during a specific sensor-malfunction sequence. Construct a Control Flow Graph (CFG) for such a logic module and evaluate the gaps in branch coverage. Propose a white-box testing strategy that ensures 100% Modified Condition/Decision Coverage (MC/DC) and justify its impact on safety-critical reliability.	5
8	A travel booking portal allows users to apply multiple promo codes, but certain combinations lead to a negative "Total Amount" due. Analyze the limitations of testing these inputs in isolation. Apply Decision Table Testing or Cause-Effect Graphing to map all valid and invalid combinations of price, promo type, and user tier. Propose an optimized test set that maximizes defect detection while minimizing redundant test cases.	5

9	A legacy database migration tool uses a deeply nested loop structure to validate records. The team reports high statement coverage but frequent "infinite loop" crashes in production. Evaluate the current white-box approach and identify why statement coverage was insufficient. Propose a Path Testing strategy focusing on independent paths and loop boundary conditions. Justify your approach using Cyclomatic Complexity metrics.	5
10	A mobile food delivery app features a "Real-time Tracker" that transitions through various states (Ordered, Preparing, Out for Delivery, Delivered). Users reported the app getting stuck when a driver cancels mid-delivery. Evaluate the current black-box strategy and apply State Transition Testing to redesign the scenarios. Justify how this technique addresses the logic gaps that ECP and BVA might miss in dynamic workflows.	5
11	A cybersecurity authentication module uses a multi-factor logic involving password strength, OTP timing, and IP location. Despite testing, "Brute Force" vulnerabilities were detected. Critically assess the integration of black-box and white-box methods used. Propose a Hybrid Testing strategy that combines Error Guessing with formal Path Analysis. Justify how this integrated approach enhances the security posture of the module.	5
12	An e-learning platform provides tiered access (Free, Basic, Premium) based on subscription dates and course types. Errors were found when a user upgrades exactly on the day of expiration. Evaluate the adequacy of the current testing criteria. Apply Equivalence Class Partitioning to the date ranges and subscription types. Propose a structured test design and justify its completeness in ensuring seamless user transitions.	5
13	A smart-home lighting controller uses a specific algorithm to dim lights based on ambient light sensors. The code contains several switch-case statements and complex Boolean expressions. Evaluate the risks of relying solely on statement coverage for this module. Apply Branch and Condition Coverage techniques to identify missing logical paths. Design an improved white-box strategy and justify how it reduces the risk of hardware-software integration failures.	5

14	A logistics management system calculates shipping costs based on weight, destination zone, and delivery speed. Testers failed to detect errors when weights were entered as zero or negative values. Analyze the failure of the current input validation testing. Apply Boundary Value Analysis (Robustness Testing) to include invalid boundaries. Justify the necessity of testing "negative scenarios" for robust system performance.	5
15	A medical imaging software processes large datasets using parallel processing loops. Inconsistent results occur only when the data size is a prime number. Evaluate the testing team's reliance on random testing for such a specialized mathematical module. Propose a structured White-Box Testing approach that incorporates Basis Path Testing. Justify your solution by explaining how it ensures mathematical accuracy and system stability.	5

1. Online Banking – Fund Transfer Failures

The failures occur because boundary values may not have been tested properly. Equivalence Partitioning (ECP) divides inputs into valid and invalid groups, while Boundary Value Analysis (BVA) tests values near the minimum and maximum limits.

Test cases should include:

- Below minimum transfer limit – invalid.
- Minimum limit – valid.
- Normal transfer amount – valid.
- Maximum limit – valid.
- Above maximum limit – invalid.
- Zero, negative, and insufficient-balance amounts – invalid.

Random testing may find unexpected errors, but it cannot guarantee boundary coverage. Therefore, structured ECP and BVA testing should be prioritized, with random testing retained as supplementary testing. This improves defect detection and reliability.

2. Healthcare Application – Defects After Deployment

The defects occurred because the testing coverage was insufficient. White-box testing examines the internal code structure.

A Control Flow Graph (CFG) represents program statements as nodes and control flow as edges. Missing paths in the graph may cause defects to remain undetected.

- Statement coverage: Executes every statement.
- Branch coverage: Tests both true and false outcomes.
- Path coverage: Tests execution paths through the program.

The team should identify all important branches, construct a CFG, and design test cases for each independent path. Complete path coverage may be impractical for complex programs, so critical paths and boundary conditions should be prioritized. This improves reliability and reduces risk.

3. E-commerce System – Price, Quantity and Discount

The existing testing failed because individual inputs were tested without checking their combinations.

The following techniques should be used:

- ECP: Divide price, quantity, and discount into valid and invalid groups.
- BVA: Test minimum, maximum, and nearby boundary values.
- Decision Table Testing: Test combinations of price, quantity, and discount conditions.

Test cases should include zero price, negative quantity, maximum discount, and invalid combinations. This approach detects more defects and improves test completeness.

4. Complex Conditions and Loops – Partial Path Coverage

Partial path coverage means some execution paths have not been tested. Incorrect CFG construction and missing independent paths may cause defects.

The testing team should:

1. Construct an accurate Control Flow Graph.
2. Identify independent execution paths.
3. Calculate Cyclomatic Complexity.
4. Test true and false outcomes of conditions.
5. Test loops for zero, one, and multiple iterations.

Complete path testing may be impossible when loops create many paths. Therefore, basis paths, critical paths, and loop boundaries should be tested to improve reliability.

5. Financial Application – Hybrid Testing

A financial application may continue to produce defects when black-box and white-box testing are not integrated effectively. Black-box testing checks external behavior, while white-box testing examines internal logic.

Technique	Purpose
ECP	**Divide inputs into valid and invalid groups**
BVA	**Test boundary values**
CFG	**Represent program flow**
Path Testing	**Test independent execution paths**
Branch Coverage	**Test true and false outcomes**

The team should test invalid financial inputs, boundary amounts, decision branches, and important execution paths. Coverage results should be reviewed to identify gaps.

Conclusion: Hybrid testing provides better coverage, detects more defects, and reduces production failures.

6. HR Payroll – Incorrect Bonus Calculation

Answer: The payroll system produces incorrect bonuses at exactly 5, 10, and 15 years of service. This indicates possible errors in boundary conditions or bonus calculation rules. BVA tests values around each boundary, while ECP divides employees into groups based on service years and performance ratings.

Boundary	Test values
5 years	**4, 5, 6**
10 years	**9, 10, 11**
15 years	**14, 15, 16**

Technique comparison:

Technique	Benefit
BVA	**Detects errors near service-year boundaries**
ECP	**Tests representative service and rating groups**

Conclusion: ECP and BVA improve bonus accuracy and reduce financial calculation defects.

7. Laboratory Temperature Control – Alarm Failure

A laboratory temperature-control system uses conditions to activate heaters, coolers, and alarms. An alarm failure may occur when a sensor-malfunction branch has not been tested. The CFG should include sensor input, temperature checks, heater control, cooler control, and alarm activation.

Test condition	Expected result
Normal temperature	**Normal operation**
High temperature	**Cooler activated**
Low temperature	**Heater activated**
Sensor malfunction	**Alarm activated**
Critical failure	**Safe response**

Modified Condition/Decision Coverage (MC/DC) requires each Boolean condition to independently affect the overall decision outcome. Test cases should demonstrate this for every safety-critical decision.

Conclusion: CFG analysis and MC/DC improve fault detection and safety-critical reliability.

8. Travel Booking – Negative Total Amount

A travel booking portal may calculate a negative total when multiple promotional codes interact incorrectly. Testing each input separately cannot identify every combination error. Decision Table Testing represents combinations of conditions and their expected results.

Input condition	Test cases
Price	**Minimum, normal and maximum**
Promo code	**Valid and invalid**
User tier	**Different user tiers**
Multiple codes	**Allowed and disallowed combinations**
Final amount	**Zero, positive and negative-result scenarios**

The system should reject invalid promotional combinations and prevent a negative payable amount. Redundant tests can be removed while maintaining coverage of important combinations.

Conclusion: Decision Table Testing improves defect detection and reduces unnecessary test cases.

9. Database Migration – Infinite Loop Crashes

High statement coverage does not guarantee that loops terminate correctly. Infinite loops may occur because of incorrect loop conditions, missing exit conditions, or improper record handling.

Test case	Expected result
Empty dataset	**Loop terminates**
One record	**Record processed correctly**
Multiple records	**All records processed**
Invalid record	**Error handled correctly**
Large dataset	**Processing terminates correctly**

The team should construct a CFG, calculate Cyclomatic Complexity, identify independent paths, and test loop boundaries. Basis Path Testing helps execute important independent paths.

Conclusion: Loop testing and basis path testing help detect infinite loops and improve migration stability.

10. Food Delivery App – Tracker Stuck During Cancellation

The failure occurs because some state transitions were not tested.

State Transition Testing checks states and transitions between them.

Current state	Event	Expected state
Ordered	**Order accepted**	**Preparing**
Preparing	**Food dispatched**	**Out for Delivery**
Out for Delivery	**Delivery completed**	**Delivered**
Out for Delivery	**Driver cancels**	**Cancellation or reassignment**

The team should test valid and invalid transitions, including cancellation and reassignment. ECP and BVA alone may miss these dynamic workflow errors. State Transition Testing improves tracking accuracy and user experience.

11. Cybersecurity – Brute Force Vulnerabilities

The vulnerabilities may exist because authentication conditions and attack scenarios were not tested adequately.

The improved hybrid strategy includes:

- Error Guessing: Test repeated incorrect passwords, expired OTPs, and invalid inputs.
- Black-box testing: Verify authentication responses and account lockout.
- Path Analysis: Examine password, OTP, and IP-location decision paths.
- Branch Coverage: Test success and failure outcomes.
- Security testing: Verify rate limiting, temporary lockout, and OTP expiration.

Combining these methods identifies security weaknesses and strengthens authentication protection.

12. E-learning Platform – Upgrade on Expiration Date

The error occurs because subscription date boundaries may not have been tested correctly.

ECP divides subscription inputs into valid categories.

Test cases should include:

- Upgrade before expiration.
- Upgrade exactly on the expiration date.
- Upgrade after expiration.
- Free to Basic upgrade.
- Basic to Premium upgrade.
- Invalid subscription type.

The expected access rights should be verified for every case. Testing date boundaries and subscription combinations ensures correct access transitions and prevents unexpected service interruptions.

13. Smart-Home Lighting – Statement Coverage Limitations

Statement coverage checks whether every statement executes, but it may miss logical errors in conditions and switch-case branches**.**

The improved strategy includes:

1. Construct a CFG.
2. Test every switch-case branch.
3. Apply branch coverage to true and false outcomes.
4. Apply condition coverage to individual Boolean conditions.
5. Test different ambient light levels and sensor failures.
6. Verify hardware responses.

Branch and condition coverage detect missing logical paths and reduce hardware-software integration failures.

Technique	Purpose
Statement Coverage	Execute every statement
Branch Coverage	Test each branch
Condition Coverage	Test each Boolean condition
CFG	Identify control flow
Integration Testing	Verify software and hardware responses

14. Logistics System – Zero and Negative Weights

The existing testing failed because invalid weight values were not tested.

Robust Boundary Value Analysis should include:

Input weight	Expected result
Negative weight	**Reject**
Zero weight	**Reject**
Smallest valid positive weight	**Accept**
Normal weight	**Accept**
Maximum weight	**Accept**
Above maximum weight	**Reject**

The system should display suitable validation messages for invalid values. Negative testing is necessary to prevent incorrect shipping charges and improve system reliability.

Conclusion: Negative testing is necessary to detect invalid inputs, prevent calculation errors, and improve system reliability.

15. Medical Imaging – Incorrect Results for Prime Data Sizes

Random testing may fail to identify errors that occur only with prime-sized datasets.

The team should use white-box testing and Basis Path Testing.

1. Construct a CFG for the parallel-processing algorithm.
2. Calculate Cyclomatic Complexity.
3. Identify independent execution paths.
4. Test prime and non-prime dataset sizes.
5. Test empty, small, and large datasets.
6. Compare parallel-processing results with expected results.

Test case	Purpose
Empty dataset	Check empty-input handling
Small dataset	Verify basic processing
Prime-sized dataset	Detect prime-size errors
Non-prime dataset	Compare processing behavior
Large dataset	Check stability and performance

This approach systematically tests important paths and data-size boundaries. It helps identify loop, indexing, and parallel-processing errors, improving mathematical accuracy and system stability.

Code of Conduct:

I _____C Prasanna(192371035)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: C Prasanna

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Context Analysis	25	Thoroughly understands the scenario with accurate interpretation of all requirements	Clearly understands the scenario with minor gaps	Basic understanding; some aspects of the scenario unclear	Poor understanding or misinterpretation of the scenario
Application of Concepts	30	Applies appropriate concepts effectively to solve complex realworld problems	Applies relevant concepts correctly with minor errors	Applies basic concepts but with limited accuracy	Fails to apply appropriate concepts
Analytical & Decision-Making Skills	20	Demonstrates strong analysis and selects optimal solutions with justification	Good analysis with reasonable solutions	Limited analysis; solutions lack justification	Poor or no analysis; incorrect decisions
Solution Design & Approach	15	Well-structured, innovative, and feasible solution approach	Logical and structured solution with minor gaps	Basic solution with limited structure or feasibility	Unclear or incorrect solution approach
Clarity, Justification & Presentation	10	Clear, well-justified explanation with strong reasoning	Clear explanation with some justification	Limited clarity and weak justification	Poorly explained with no justification